LESSON 7.2

Represent Rational Numbers and Their Opposites on the Number Line

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can place rational numbers, including fractions and decimals, on a number line.

WHY THIS MATTERS Arihi planted a flower. The height and depth of the plant are shown. How do the height of the plant and the depth of the roots compare?

REMEMBER

Represent Rational Numbers and Their Opposites on the Number Line

Rational Numbers on the Number Line A rational number can be written as a ratio of two integers with a nonzero denominator. Integers and terminating or repeating decimals are rational. A rational number that is not an integer lies between two neighboring integers.

1. Find the two consecutive integers the rational number falls between.
2. Divide the segment between integers into equal parts matching the denominator.
3. Negative numbers move to the left of zero; positive numbers move to the right

Words: **Rational number *(Número racional)*** · **Fraction *(Fracción)*** · **Decimal *(Decimal)*** · **Number line *(Recta numérica)*** · **Equivalent *(Equivalente)***

1 **Circle the letter of the best answer. Show how you know.**

What is the opposite of -3/4?

(A) -3/4 (B) 0

(C) 3/4 (D) 4/3

HINT 'Opposite' means the mirror image across zero — same distance, other side.

Rewrite each fraction _____

Work it out _____

Simplify _____

1. **Same size pieces** *Common denominator first.*
2. **Operate** *Only then add or subtract.*
3. **Simplify** *Divide out common factors.*

2 **Circle the letter of the best answer. Show how you know.**

What is the opposite of −6?

(A) 0
(B) 6
(C) −6
(D) −12

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

On a number line, which number is farther to the LEFT: −8 or −3?

(A) −3
(B) They are the same
(C) −8
(D) Neither is on the line

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Which integer is the LEAST: −3, 5, −8, 2, 0?

(A) 0
(B) −3
(C) 5
(D) −8

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A point sits 5 units to the LEFT of 0 on a number line. What number is it?

(A) 0
(B) −5
(C) 5
(D) −10

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

The boat ramp at Miller Lake has a full mark painted on it. When the water sits exactly on that mark, the ramp keeper writes 0 — the mark IS zero. Above the mark is positive, below it is negative. Last week she wrote: Monday +0.75 ft, Tuesday −1.25 ft, Wednesday −0.5 ft, Thursday +1.5 ft. Where does −1.25 fall on the number line?

(A) Between −1 and −2
(B) Between −1 and 0
(C) Between 1 and 2
(D) Exactly at −1

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How did you decide exactly where to place rational numbers that fall between integers?

SENTENCE STARTER To plot ___, I found the integers it falls between, which are ___ and ___. Then I estimated ___ of the way between them because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 7.2

Represent Rational Numbers and Their Opposites on the Number Line

Set B · Version B · Second form · core practice

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Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can place rational numbers, including fractions and decimals, on a number line.

WHY THIS MATTERS Arihi planted a flower. The height and depth of the plant are shown. How do the height of the plant and the depth of the roots compare?

PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each decimal trail marker to its equal fraction.

_____ 0.5

_____ 0.25

_____ -0.5

_____ 0.2

(A) 1/2

(B) 1/4

(C) -1/2

(D) 1/5

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

The boat ramp at Miller Lake has a full mark painted on it. When the water sits exactly on that mark, the ramp keeper writes 0 — the mark IS zero. Above the mark is positive, below it is negative. Last week she wrote: Monday +0.75 ft, Tuesday −1.25 ft, Wednesday −0.5 ft, Thursday +1.5 ft. Which day had the water level closest to the painted mark?

(A) Wednesday at −0.5 ft

(B) Monday at +0.75 ft

(C) Tuesday at −1.25 ft

(D) Thursday at +1.5 ft

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

The boat ramp at Miller Lake has a full mark painted on it. When the water sits exactly on that mark, the ramp keeper writes 0 — the mark IS zero. Above the mark is positive, below it is negative. Last week she wrote: Monday +0.75 ft, Tuesday −1.25 ft, Wednesday −0.5 ft, Thursday +1.5 ft. Why are rational numbers more useful here than integers alone?

(A) They let the scientist record part-of-a-foot changes like 0.75 and −1.25

(B) They are easier to add

(C) Integers cannot be negative

(D) They make the lake bigger

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

The first trail marker sits between 0 and 1. Which rational number belongs there?

(A) -1/2

(B) 1.5

(C) 2

(D) 3/4

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Order these three trail markers from least to greatest: -1.5, 0.5, -0.5.

(A) -1.5, -0.5, 0.5

(B) -1.5, 0.5, -0.5

(C) -0.5, -1.5, 0.5

(D) 0.5, -0.5, -1.5

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

On the trail map, -3/4 sits to the LEFT of which marker?

(A) -7/8
(B) -2
(C) -1/4
(D) -1

Rewrite each fraction _____

Work it out _____

Simplify _____

EXPLAIN YOUR THINKING

How did you decide exactly where to place rational numbers that fall between integers?

_____

_____

_____

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help